

```
"""
```

```
Custom exceptions for the Embedded Sensor Network Data Aggregation Hub.
```

```
NOTE: This is a minimal placeholder so that ``src/sensors.py`` and its  
test suite can be developed and run independently. Replace this file  
with the version owned by the teammate responsible for  
``src/exceptions.py``; the public interface (constructor signature and  
stored attributes) is matched to the shared spec so no other code  
should need to change when it is swapped in.
```

```
"""
```

```
from datetime import datetime
```

```
class SensorError(Exception):
```

```
    """Base class for all sensor-related exceptions in this project."""
```

```
    def __init__(self, sensor_id: str, detail: str):
```

```
        """
```

```
        Initialize a sensor error.
```

```
        Args:
```

```
            sensor_id: The identifier of the sensor that raised the error.
```

```
            detail: A human-readable explanation of what went wrong.
```

```
        """
```

```
        self.sensor_id = sensor_id
```

```
        self.detail = detail
```

```
        self.timestamp = datetime.now()
```

```
        super().__init__(f"[{sensor_id}] {detail}")
```

```
class SensorOfflineError(SensorError):
```

```
    """Raised when a sensor is unreachable or fails a self-test."""
```

```
class CalibrationError(SensorError):
```

```
    """Raised when a calibration offset is invalid for a sensor."""
```

```
class CommunicationTimeoutError(SensorError):
```

```
    """Raised when communication with a sensor times out."""
```

